HDOC Day-65 Activity

Tower of Hanoi Using Recursion

Student Name	Dhruv Pandey
SAP ID	590037660
Batch	15
Course / Activity	HDOC Day-65

Aim

To write a recursive C program to solve the **Tower of Hanoi** problem for n disks and display every move required to transfer all disks from the source rod to the destination rod.

Problem Rules

- There are three rods: Source (A), Auxiliary (B), and Destination (C).
- Initially, all n disks are placed on the source rod, with the largest disk at the bottom.
- Only one disk can be moved at a time.
- Only the top disk of a rod can be moved.
- A larger disk can never be placed on top of a smaller disk.

Recursive Idea

To move n disks from A to C, first move the top n-1 disks from A to B, then move disk n from A to C, and finally move the n-1 disks from B to C. The same procedure is recursively repeated for smaller groups of disks.

Algorithm

1. Read the number of disks n.
2. Call **tower_of_hanoi(n, 'A', 'C', 'B')**.
3. Base case: if n == 1, move disk 1 directly from the source rod to the destination rod and return.
4. Recursively move n-1 disks from source to auxiliary using destination as temporary storage.
5. Display the move of disk n from source to destination.
6. Recursively move n-1 disks from auxiliary to destination using source as temporary storage.
7. Continue until all recursive calls finish.

C Program

```
#include <stdio.h>

void tower_of_hanoi(int n, char source,
                    char destination, char auxiliary)
{
    if (n == 1)
    {
        printf("Move disk 1 from %c to %c\n",
               source, destination);
        return;
    }

    tower_of_hanoi(n - 1, source,
                   auxiliary, destination);

    printf("Move disk %d from %c to %c\n",
           n, source, destination);

    tower_of_hanoi(n - 1, auxiliary,
                   destination, source);
}

int main()
{
    int n;

    printf("Enter number of disks: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Number of disks must be positive.");
        return 0;
    }

    tower_of_hanoi(n, 'A', 'C', 'B');

    return 0;
}
```

Function Parameters

n is the number of disks being moved. **source** is the rod from which the disks start, **destination** is the target rod, and **auxiliary** is the temporary rod used during the recursive process.

Dry Run for n = 3

Move No.	Operation
1	Move disk 1 from A to C
2	Move disk 2 from A to B
3	Move disk 1 from C to B
4	Move disk 3 from A to C
5	Move disk 1 from B to A
6	Move disk 2 from B to C
7	Move disk 1 from A to C

For three disks, all disks are transferred from rod A to rod C in **7 moves**, while following all Tower of Hanoi rules.

Recursion Breakdown

Call	Action
Hanoi(3, A, C, B)	Move 2 disks A -> B, move disk 3 A -> C, move 2 disks B -> C
Hanoi(2, A, B, C)	Move 1 disk A -> C, move disk 2 A -> B, move 1 disk C -> B
Hanoi(1, ...)	Base case: directly move the single disk

Number of Moves

The minimum number of moves required for n disks is $\mathbf{2^n - 1}$. For example, n = 1 requires 1 move, n = 2 requires 3 moves, n = 3 requires 7 moves, and n = 4 requires 15 moves.

Complexity

Time Complexity: $O(2^n)$ because the number of recursive moves grows exponentially and exactly $2^n - 1$ moves are displayed.

Space Complexity: O(n) because the maximum recursion depth is n.

Result

The program successfully solves the Tower of Hanoi problem using recursion and displays every valid move required to transfer all n disks from the source rod to the destination rod.

Execution Example

```
Enter number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```